

the eigenvectors such that D is orthogonal, that is, $D^{-1} = D^T$. We therefore obtain the Jordan form decomposition of Q as

$$Q = D\hat{Q}D^T \quad (2.106)$$

where $\hat{Q}$ is the diagonal matrix of the eigenvalues of Q . That is,

$$\hat{Q} = \text{diag}(\mu_1^2, \dots, \mu_n^2) \quad (2.107)$$

Now we define the random vector v as $v = D^{-1}w$, so that $w = Dv$. Therefore,

$$\begin{aligned} E(vv^T) &= E(D^T w w^T D) \\ &= D^T Q D \\ &= \hat{Q} \\ &= \text{diag}(\mu_1^2, \dots, \mu_n^2) \end{aligned} \quad (2.108)$$

This shows how we can generate an n -element random vector w with a covariance matrix of Q . The algorithm is given as follows.

Correlated noise simulation

1. Find the eigenvalues of Q , and denote them as $\mu_1^2, \dots, \mu_n^2$
2. Find the eigenvectors of Q , and denote them as $d_1, \dots, d_n$, such that

$$\begin{aligned} D &= [d_1 \ \dots \ d_n] \\ D^{-1} &= D^T \end{aligned} \quad (2.109)$$

3. For $i = 1, \dots, n$ compute the random variable $v_i = \mu_i r_i$, where each r_i is an independent random number with a variance of 1 (unity variance).
4. Set $w = Dv$.

2.8 SUMMARY

In this chapter, we have reviewed the basic concepts of probability, random variables, and stochastic processes. The probability of some event occurring is simply and intuitively defined as the number of times the event occurs divided by the number of chances the event has to occur. A random variable (RV) is a variable whose value is not certain, but is governed by the laws of probability. For example, your score on the test for this chapter is not deterministic, but is a random variable. Your *actual* score, after you take the test, will be a specific, deterministic number. But *before* you take the test, you do not know what you will get on the test. You may suppose that you will probably get between 80% and 90% if you have a decent understanding of the material, but your actual score will be determined by random events such as your health, how well you sleep the night before, what topics the instructor decides to cover on the test versus what topics you study, what the traffic was like on the way to school, the mood of the instructor when she grades the test, and so on. A stochastic process is a random variable that changes with time,

such as your performance on all of the quizzes and homework assignments for this course. The expected value of your test grades may be constant throughout the duration of the course if you are a consistent person, or it may increase if you tend to study harder as the course progresses, or it may decrease if you tend to study less as the course progresses. Probability, random variables, stochastic processes, and related topics form a huge area of study that we have only touched on in this chapter. Additional information on these topics can be found in many textbooks, including [Pap02, Pee01]. A study of these topics will allow a student to delve into many practical engineering subjects, including control and estimation theory, signal processing, and communications theory.

PROBLEMS

Written exercises

2.1 What is the 0th moment of an RV? What is the 0th central moment of an RV?

2.2 Suppose a deck of 52 cards is randomly divided into four piles of 13 cards each. Find the probability that each pile contains exactly one ace [Gre01].

2.3 Determine the value of a in the function

$$f_X(x) = \begin{cases} ax(1-x) & x \in [0, 1] \\ 0 & \text{otherwise} \end{cases}$$

so that $f_X(x)$ is a valid probability density function [Lie67].

2.4 Determine the value of a in the function

$$f_X(x) = \frac{a}{e^x + e^{-x}}$$

so that $f_X(x)$ is a valid probability density function. What is the probability that $|X| \leq 1$?

2.5 The probability density function of an exponentially distributed random variable is defined as follows.

$$f_X(x) = \begin{cases} ae^{-ax} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

where $a \geq 0$.

- a) Find the probability distribution function of an exponentially distributed random variable.
- b) Find the mean of an exponentially distributed random variable.
- c) Find the second moment of an exponentially distributed random variable.
- d) Find the variance of an exponentially distributed random variable.
- e) What is the probability that an exponentially distributed random variable takes on a value within one standard deviation of its mean?

2.6 Derive an expression for the skew of a random variable as a function of its first, second, and third moments.

2.7 Consider the following probability density function:

$$f_X(x) = \frac{ab}{b^2 + x^2}, \quad b > 0$$

- a) Determine the value of a in the so that $f_X(x)$ is a valid probability density function. (The correct value of a makes $f_X(x)$ a Cauchy pdf.)
- b) Find the mean of a Cauchy random variable.

2.8 Consider two zero-mean uncorrelated random variables W and V with standard deviations σ_w and σ_v , respectively. What is the standard deviation of the random variable $X = W + V$?

2.9 Consider two scalar RVs X and Y .

- a) Prove that if X and Y are independent, then their correlation coefficient $\rho = 0$.
- b) Find an example of two RVs that are not independent but that have a correlation coefficient of zero.
- c) Prove that if Y is a linear function of X then $\rho = \pm 1$.

2.10 Consider the following function [Lie67].

$$f_{XY}(x, y) = \begin{cases} ae^{-2x}e^{-3y} & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

- a) Find the value of a so that $f_{XY}(x, y)$ is a valid joint probability density function.
- b) Calculate $\bar{x}$ and $\bar{y}$.
- c) Calculate $E(X^2)$, $E(Y^2)$, and $E(XY)$.
- d) Calculate the autocorrelation matrix of the random vector $[X \ Y]^T$.
- e) Calculate the variance σ_x^2 , the variance σ_y^2 , and the covariance C_{XY} .
- f) Calculate the autocovariance matrix of the random vector $[X \ Y]^T$.
- g) Calculate the correlation coefficient between X and Y .

2.11 A stochastic process has the autocorrelation $R_X(\tau) = Ae^{-k|\tau|}$, where A and k are positive constants.

- a) What is the power spectrum of the stochastic process?
- b) What is the total power of the stochastic process?
- c) What value of k results in half of the total power residing in frequencies less than 1 Hz?

2.12 Suppose X is a random variable, and $Y(t) = X \cos t$ is a stochastic process.

- a) Find the expected value of $Y(t)$.
- b) Find $A[Y(t)]$, the time average of $Y(t)$.
- c) Under what condition is $\bar{y}(t) = A[Y(t)]$?

2.13 Consider the equation $Z = X + V$. The pdf's of X and B are given in Figure 2.5.

- a) Plot the pdf of $(Z|X)$ as a function of X for $Z = 0.5$.

- b) Given $Z = 0.5$, what is conditional expectation of X ? What is the most probable value of X ? What is the median value of X ?

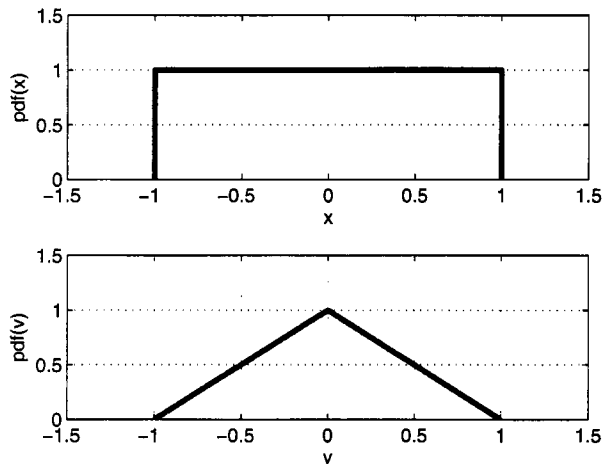


Figure 2.5 pdf's for Problem 2.13 [Sch73].

- 2.14** The temperature at noon in London is a stochastic process. Is it ergodic?

Computer exercises

2.15 Generate $N = 50$ independent random numbers, each uniformly distributed between 0 and 1. Plot a histogram of the random numbers using 10 bins. What is the sample mean and standard deviation of the numbers that you generated? What would you expect to see for the mean and standard deviation (i.e., what are the theoretical mean and standard deviation)? Repeat for $N = 500$ and $N = 5,000$ random numbers. What changes in the histogram do you see as N increases?

2.16 Generate 10,000 samples of $(x_1 + x_2)/2$, where each x_i is a random number uniformly distributed on $[-1/2, +1/2]$. Plot the 50-bin histogram. Repeat for $(x_1 + x_2 + x_3 + x_4)/4$. Describe the difference between the two histograms.

